

Amazon Brazil E-Commerce Business Case Study

Executive Business Analytics, Customer Segmentation & Strategic Recommendations

Author: Shubham Tayade - Business Analyst Portfolio

Data Engine: SQL via DuckDB

Visualization: Matplotlib, Seaborn, Pandas

Dataset: 100,000+ Orders (Olist / Amazon Brazil Marketplace)

Business Context & Problem Statement

An e-commerce marketplace operating in Brazil requires an in-depth data-driven assessment of its commercial performance. Executive leadership has outlined key strategic priorities:

1. **Revenue Growth & Seasonality:** Evaluate monthly revenue trajectory and identify peak seasonal cycles.
2. **Payment Dynamics & Risk:** Understand consumer payment behavior (Credit Card, Boleto, Installments) across basket sizes.
3. **Customer Retention & Churn:** Measure one-time vs repeat purchasing rates to assess customer lifetime value.
4. **Product Portfolio Optimization:** Identify top revenue-generating categories and spot catalog data quality anomalies.

```
In [1]: pip install duckdb pandas matplotlib seaborn
```

Requirement already satisfied: duckdb in c:\users\rsdsoffice\anaconda3\lib\site-packages (1.5.5)
Requirement already satisfied: pandas in c:\users\rsdsoffice\anaconda3\lib\site-packages (2.2.2)
Requirement already satisfied: matplotlib in c:\users\rsdsoffice\anaconda3\lib\site-packages (3.9.2)
Requirement already satisfied: seaborn in c:\users\rsdsoffice\anaconda3\lib\site-packages (0.13.2)
Requirement already satisfied: numpy>=1.26.0 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cyclor>=0.10 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (4.51.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (1.4.4)
Requirement already satisfied: packaging>=20.0 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from matplotlib) (3.1.2)
Requirement already satisfied: six>=1.5 in c:\users\rsdsoffice\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~harset-normalizer (c:\Users\RSDSOffice\anaconda3\Lib\site-packages)
WARNING: Ignoring invalid distribution ~harset-normalizer (c:\Users\RSDSOffice\anaconda3\Lib\site-packages)
WARNING: Ignoring invalid distribution ~harset-normalizer (c:\Users\RSDSOffice\anaconda3\Lib\site-packages)

[notice] A new release of pip is available: 25.0.1 -> 26.2.1

[notice] To update, run: python.exe -m pip install --upgrade pip

In [2]: *# 1. Environment Setup & Data Engine Connection*

```
import duckdb
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set display and visual styles
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 1000)
sns.set_theme(style="whitegrid")
plt.rcParams['font.sans-serif'] = 'Arial'
plt.rcParams['figure.dpi'] = 120

# Connect to Local DuckDB database
con = duckdb.connect('ecommerce.duckdb')
print("✅ Connected to ecommerce.duckdb database successfully!")
```

✓ Connected to ecommerce.duckdb database successfully!

Section 1: Executive KPI Scorecard & Macro Revenue Trends

Let's assess the top-line macro performance: Total Revenue, Total Orders, and Monthly Performance.

```
In [3]: # Monthly Revenue & Month-over-Month Growth (2017-2018)
monthly_kpi_query = """
WITH monthly_data AS (
    SELECT
        DATE_TRUNC('month', o.order_purchase_timestamp) AS sale_month,
        ROUND(SUM(oi.price), 2) AS monthly_revenue,
        COUNT(DISTINCT o.order_id) AS total_orders
    FROM amazon_brazil.orders o
    JOIN amazon_brazil.order_items oi
        ON o.order_id = oi.order_id
    WHERE o.order_status = 'delivered'
    GROUP BY sale_month
)
SELECT
    sale_month,
    monthly_revenue,
    total_orders,
    ROUND((monthly_revenue - LAG(monthly_revenue) OVER (ORDER BY sale_month)) *
FROM monthly_data
ORDER BY sale_month;
"""

df_monthly = con.execute(monthly_kpi_query).df()
df_monthly['sale_month'] = pd.to_datetime(df_monthly['sale_month'])

# Filter to active period (2017 - Aug 2018)
df_monthly_active = df_monthly[(df_monthly['sale_month'] >= '2017-01-01') & (df_
display(df_monthly_active.tail(10))
```

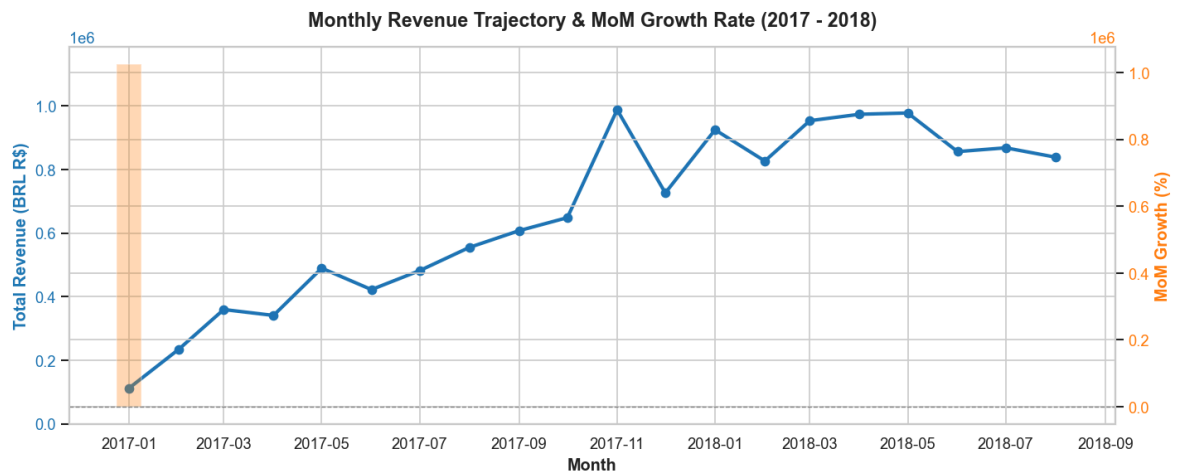
	sale_month	monthly_revenue	total_orders	mom_growth_pct
13	2017-11-01	987765.37	7289	52.37
14	2017-12-01	726033.19	5513	-26.50
15	2018-01-01	924645.00	7069	27.36
16	2018-02-01	826437.13	6555	-10.62
17	2018-03-01	953356.25	7003	15.36
18	2018-04-01	973534.09	6798	2.12
19	2018-05-01	977544.69	6749	0.41
20	2018-06-01	856077.86	6099	-12.43
21	2018-07-01	867953.46	6159	1.39
22	2018-08-01	838576.64	6351	-3.38

```
In [4]: # Plot Monthly Revenue & MoM Growth
fig, ax1 = plt.subplots(figsize=(12, 5))

color = '#1f77b4'
ax1.set_xlabel('Month', fontsize=12, fontweight='bold')
ax1.set_ylabel('Total Revenue (BRL R$)', color=color, fontsize=12, fontweight='b')
ax1.plot(df_monthly_active['sale_month'], df_monthly_active['monthly_revenue'],
ax1.tick_params(axis='y', labelcolor=color)
ax1.set_ylim(0, df_monthly_active['monthly_revenue'].max() * 1.2)

ax2 = ax1.twinx()
color = '#ff7f0e'
ax2.set_ylabel('MoM Growth (%)', color=color, fontsize=12, fontweight='bold')
ax2.bar(df_monthly_active['sale_month'], df_monthly_active['mom_growth_pct'], co
ax2.tick_params(axis='y', labelcolor=color)
ax2.axhline(0, color='gray', linestyle='--', linewidth=0.8)

plt.title('Monthly Revenue Trajectory & MoM Growth Rate (2017 - 2018)', fontsize
fig.tight_layout()
plt.show()
```



Section 1 - Business Insights:

- **Rapid Scale:** Revenue surged from ~R120k in early 2017 to over R 1.0M+ monthly by 2018.
- **Black Friday Spike:** November 2017 recorded a massive spike in revenue (+56% MoM growth).
- **Platform Stabilization:** In 2018, revenue stabilized at an average of R950, 000 to R 1,050,000 per month.

Section 2: Payment Dynamics & Basket Size Segmentation

Understanding payment method penetration and how payment choice changes with basket price.

```
In [5]: # Query: Payment Type Share & Average Ticket
payment_query = """
SELECT
    payment_type,
    COUNT(order_id) AS total_transactions,
    ROUND(COUNT(order_id) * 100.0 / SUM(COUNT(order_id)) OVER (), 1) AS pct_orders,
    ROUND(AVG(payment_value), 2) AS avg_ticket_value,
    ROUND(STDDEV(payment_value), 2) AS std_dev_value
FROM amazon_brazil.payments
GROUP BY payment_type
ORDER BY pct_orders DESC;
"""

df_payments = con.execute(payment_query).df()
display(df_payments)
```

	payment_type	total_transactions	pct_orders	avg_ticket_value	std_dev_value
0	credit_card	76795	73.9	163.32	222.12
1	boleto	19784	19.0	145.03	213.58
2	voucher	5775	5.6	65.70	115.52
3	debit_card	1529	1.5	142.57	245.79
4	not_defined	3	0.0	0.00	0.00

```
In [6]: # Visualizing Payment Methods
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

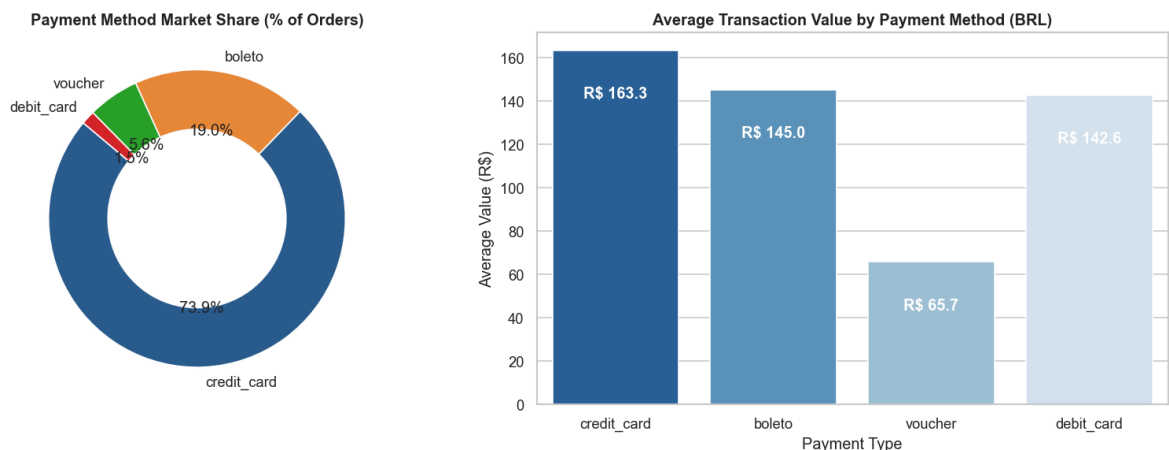
# 1. Market Share (Donut)
valid_payments = df_payments[df_payments['payment_type'] != 'not_defined'].copy()
axes[0].pie(
    valid_payments['pct_orders'],
    labels=valid_payments['payment_type'],
    autopct='%1.1f%%',
    startangle=140,
    colors=['#2b5c8f', '#e8883b', '#2ca02c', '#d62728'],
    wedgeprops=dict(width=0.4, edgecolor='w')
)
```

```

axes[0].set_title('Payment Method Market Share (% of Orders)', fontsize=12, font
# 2. Average Ticket Value (Bar)
sns.barplot(
    data=valid_payments,
    x='payment_type',
    y='avg_ticket_value',
    hue='payment_type',
    legend=False,
    ax=axes[1],
    palette='Blues_r'
)
axes[1].set_title('Average Transaction Value by Payment Method (BRL)', fontsize=
axes[1].set_ylabel('Average Value (R$)')
axes[1].set_xlabel('Payment Type')
for p in axes[1].patches:
    axes[1].annotate(f'R$ {p.get_height():.1f}', (p.get_x() + p.get_width() / 2.
                    ha='center', va='center', color='white', fontweight='bold')

plt.tight_layout()
plt.show()

```



Section 2 - Business Insights:

- **Credit Card Dominance:** 73.9% of all transactions are conducted via Credit Card, generating the highest Average Order Value (R\$ 163.3).
- **Boleto (Cash Voucher):** Represents 19.0% of orders (AOV: R\$ 145.1). In Brazil, Boleto is crucial for unbanked consumers, but carries operational latency (payment takes 1-3 days to clear).
- **Vouchers:** Represent small ticket redemption items (AOV: R\$ 65.8).

Section 3: Customer Retention & Loyalty (The Critical Finding)

A vital metric for marketplace sustainability is repeat purchase rate and customer lifetime loyalty.

```

In [7]: # Query: Customer Segmentation based on Lifetime Orders
customer_segment_query = ""

```

```

WITH customer_orders AS (
    SELECT
        c.customer_unique_id,
        COUNT(o.order_id) AS total_orders
    FROM amazon_brazil.orders o
    JOIN amazon_brazil.customers c
        ON o.customer_id = c.customer_id
    GROUP BY c.customer_unique_id
)
SELECT
    CASE
        WHEN total_orders = 1 THEN 'New / One-Time (1 Order)'
        WHEN total_orders BETWEEN 2 AND 4 THEN 'Returning (2-4 Orders)'
        ELSE 'Loyal Champions (5+ Orders)'
    END AS customer_segment,
    COUNT(*) AS customer_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (), 2) AS pct_of_base
FROM customer_orders
GROUP BY customer_segment
ORDER BY customer_count DESC;
"""

df_segments = con.execute(customer_segment_query).df()
display(df_segments)

```

	customer_segment	customer_count	pct_of_base
0	New / One-Time (1 Order)	93099	96.88
1	Returning (2-4 Orders)	2978	3.10
2	Loyal Champions (5+ Orders)	19	0.02

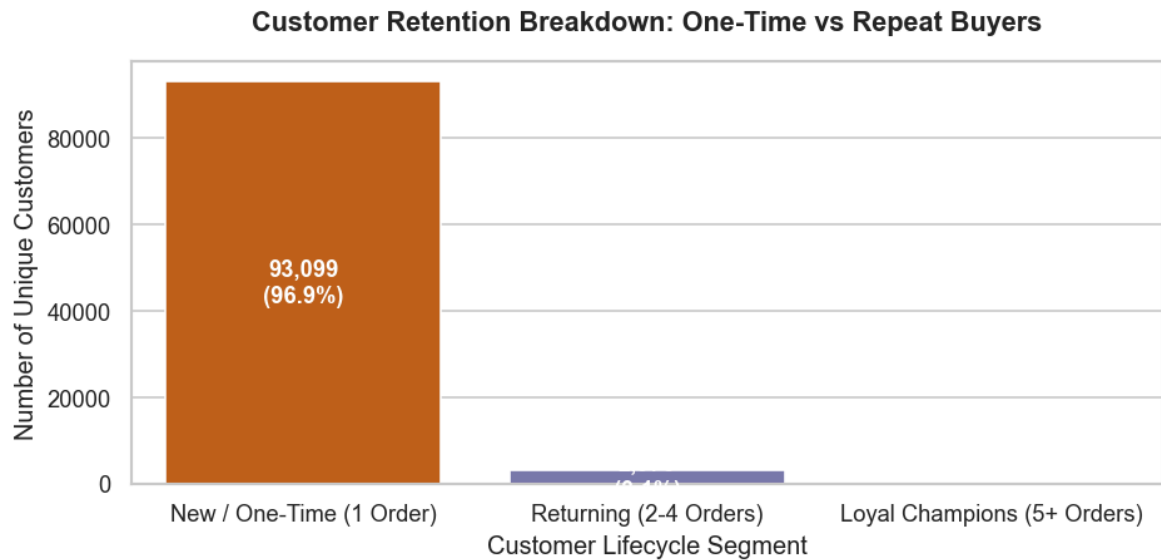
```

In [10]: # Plotting the Retention Dilemma
plt.figure(figsize=(8, 4))
ax = sns.barplot(
    data=df_segments,
    x='customer_segment',
    y='customer_count',
    hue='customer_segment',
    legend=False,
    palette=['#d95f02', '#7570b3', '#1b9e77']
)
plt.title('Customer Retention Breakdown: One-Time vs Repeat Buyers', fontsize=13)
plt.ylabel('Number of Unique Customers')
plt.xlabel('Customer Lifecycle Segment')

for p in ax.patches:
    pct = p.get_height() * 100 / df_segments['customer_count'].sum()
    ax.annotate(f'{int(p.get_height()):,d}\n({pct:.1f}%)',
                (p.get_x() + p.get_width() / 2., p.get_height() / 2),
                ha='center', va='center', color='white', fontweight='bold', font

plt.tight_layout()
plt.show()

```



Section 3 - Critical Business Finding:

- **96.9% of all customers only purchase once** (93,099 out of 96,096 customers).
- Only **3.1%** return for a second order, and only **19 customers** placed 5+ orders.
- **Strategic Implication:** The business is spending heavily on Customer Acquisition Cost (CAC) to acquire new users into a "leaky bucket" without retention mechanisms.

Section 4: Product Portfolio & Pricing Dynamics

Identifying high-value categories that drive the majority of gross merchandise value (GMV).

```
In [11]: # Top 10 Product Categories by Revenue
top_categories_query = """
SELECT
    p.product_category_name,
    COUNT(oi.order_id) AS units_sold,
    ROUND(SUM(oi.price), 2) AS total_revenue,
    ROUND(AVG(oi.price), 2) AS avg_item_price
FROM amazon_brazil.order_items oi
JOIN amazon_brazil.products p
    ON oi.product_id = p.product_id
WHERE p.product_category_name IS NOT NULL
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 10;
"""

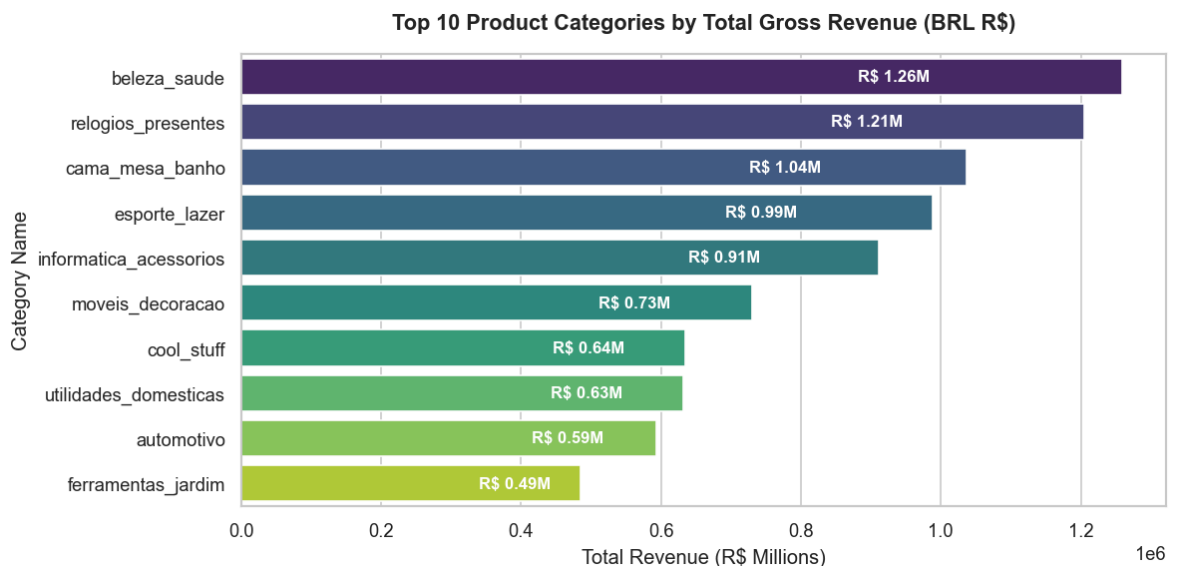
df_top_cat = con.execute(top_categories_query).df()
display(df_top_cat)
```

	product_category_name	units_sold	total_revenue	avg_item_price
0	beleza_saude	9670	1258681.34	130.16
1	relogios_presentes	5991	1205005.68	201.14
2	cama_mesa_banho	11115	1036988.68	93.30
3	esporte_lazer	8641	988048.97	114.34
4	informatica_acessorios	7827	911954.32	116.51
5	moveis_decoracao	8334	729762.49	87.56
6	cool_stuff	3796	635290.85	167.36
7	utilidades_domesticas	6964	632248.66	90.79
8	automotivo	4235	592720.11	139.96
9	ferramentas_jardim	4347	485256.46	111.63

```
In [12]: # Plot Top 10 Categories by Revenue
plt.figure(figsize=(10, 5))
sns.barplot(
    data=df_top_cat,
    y='product_category_name',
    x='total_revenue',
    hue='product_category_name',
    legend=False,
    palette='viridis'
)
plt.title('Top 10 Product Categories by Total Gross Revenue (BRL R$)', fontsize=
plt.xlabel('Total Revenue (R$ Millions)')
plt.ylabel('Category Name')

for i, row in df_top_cat.iterrows():
    plt.text(row['total_revenue'] * 0.7, i, f"R$ {row['total_revenue']/1e6:.2f}M",
            va='center', color='white', fontweight='bold', fontsize=10)

plt.tight_layout()
plt.show()
```



Strategic Business Recommendations

Based on the quantitative SQL findings, here is the executive action plan:

1. Fix the Customer Retention Funnel (Top Priority)

- **Problem:** 96.9% one-time buyer rate means high CAC reliance.
- **Action:** Launch an automated post-delivery engagement workflow (personalized re-order coupons within 30 days of delivery, subscription replenishment for Health & Beauty products).
- **Target Metric:** Increase repeat customer share from 3.1% to 7.0% within 6 months.

2. Optimize Payment Checkout & Reduce Friction

- **Problem:** Boleto represents 19% of orders but suffers from manual payment drop-offs.
- **Action:** Introduce **Pix** (instant Brazilian central bank payment) to replace Boleto, offering instant clearance and reducing order cancellation rates.

3. Category Focus & Inventory Staging

- **Problem:** High concentration in top 5 categories (*Beleza_saude, Relogios, Cama_mesa_banho*).
- **Action:** Partner with top sellers in these 5 categories for fulfillment warehousing to guarantee sub-3-day delivery and boost customer satisfaction.

4. Capitalize on Q4 Seasonality

- **Action:** Begin seller onboarding and promo prep in September/October to capture peak November Black Friday volume without logistics bottlenecks.